Energy-Efficient LED Management in Smart Parking Systems: A Reinforcement Learning Approach

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***Abstract*—Smart parking systems are a cornerstone of smart city mobility, but their LED-based guidance systems lead to significant energy consumption. While adaptive strategies exist, they often rely on fixed rules and do not optimize vehicle flow. This paper introduces a novel hybrid management strategy that jointly optimizes operational efficiency and energy consumption. We propose a Reinforcement Learning (RL) agent that learns an optimal policy for vehicle load balancing across multiple floors. This strategic assignment is coupled with a deterministic rule: a floor's LED guidance system is activated only when its occupancy exceeds a 50% threshold. Through a simulation-based comparative analysis, we demonstrate the synergy of this method. Experimental results show that our RL-based strategy reduces total energy consumption by 81.8% compared to traditional systems, while optimizing operational efficiency through intelligent load balancing. These findings establish that a proactive, learning-based assignment policy is a highly effective solution for creating parking systems that are not only smarter but also more sustainable.**

***Index Terms*—mart parking systems, Reinforcement Learning, Energy Efficiency, LED management, Digital Twin, IoT Optimization.**

I. Introduction

The unprecedented rate of global urbanization, with over 68% of the world's population expected to live in cities by 2050, has created complex challenges for urban infrastructure management [1]. Vehicle ownership continues to rise, intensifying parking demand and creating significant issues of congestion and energy consumption. In response, IoT-based smart parking systems have become a critical component of smart cities, aiming to provide real-time information to optimize space utilization [2].

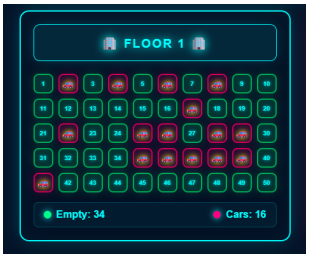
These systems introduce a significant operational challenge: the high energy consumption of their LED guidance systems. The traditional strategy, illustrated in Figure 1(a), keeps all LEDs continuously powered (50 active LEDs for 16 cars in this example), leading to structural energy waste. An improvement, the adaptive strategy shown in Figure 1(b), reduces consumption by illuminating only vacant (green) spots (34 active LEDs for 16 cars). However, this approach remains inefficient during low-traffic periods.

This paper introduces a hybrid, intelligent strategy that couples a deterministic rule with a proactive Reinforcement Learning (RL) agent. The system operates in two distinct modes: a zero-power 'standby mode' during low occupancy ($\leq$50%), as a driver has a high probability of finding a spot simply by traversing the floor (Figure 1(c)), and an 'active guidance mode' that engages only when occupancy exceeds this critical threshold (Figure 1(d)). The RL agent's task is to learn an optimal load-balancing policy to keep the system in the power-saving standby mode for as long as possible. This paper presents the modeling, simulation, and comparative analysis of this approach, demonstrating its superiority over baseline strategies.

A. Problem Statement and Motivation

The energy consumption challenge in smart parking systems is increasingly critical. The traditional approach, while functional, leads to structural energy waste. Our initial reflection led us to consider simpler adaptive strategies, such as a "Green/Off" approach that improves both energy savings and user visibility. Furthermore, we observed that light-based guidance is superfluous when the parking facility is not busy. This led to a second hypothesis: activating the guidance only when a floor's occupancy exceeds a 50% threshold.

However, this rule alone is insufficient. Without intelligent flow management, one floor can fill up prematurely and activate its lighting system, while other floors remain empty. The motivation of this work is therefore to solve this optimization problem by creating a balance between vehicle flow management and energy management.



(a) Parking Strategy 1

(b) Parking Strategy 2

(c) Parking Strategy 3

(d) Parking Strategy 4

Fig. 1: Comparison of LED management strategies.

B. Research Objectives

The primary objective of this research is to develop and evaluate an intelligent parking management strategy that jointly optimizes operational efficiency and energy consumption.

Our specific goals are:

1) To model and compare three LED management strategies: a traditional one (baseline), an adaptive one (benchmark), and an intelligent approach based on Reinforcement Learning.
2) To design and implement a Reinforcement Learning agent capable of learning an optimal load-balancing policy.
3) To quantitatively validate through simulation the performance of our approach on key energy-saving and operational efficiency metrics.

C. Proposed Methodology Overview

Our research proposes an innovative hybrid approach. It couples a simple operational rule with strategic intelligence:

- Operational Rule: The light guidance for a floor (green LEDs for vacant spots) is only activated if its occupancy rate exceeds 50%.
- Strategic Intelligence: A Reinforcement Learning (RL) agent is trained to assign incoming vehicles. Its goal is to learn a load-balancing policy that delays as long as possible the moment any floor reaches the activation threshold.

We exclude from this study specific use cases such as reserved parking spots, which will be the subject of future work.

D. Contributions and Significance

This research makes several significant contributions, that are:

- Novel Joint Optimization Strategy: Introduction of an approach that directly links vehicle assignment policy to the infrastructure's energy-saving objective.
- Empirical Validation: A simulation-based comparative analysis that quantifies the gains of our RL approach against baseline strategies.
- Demonstration of Synergy: We show how RL-based load balancing acts as a lever to maximize the effectiveness of an energy-saving rule.

The significance of this work lies in its proposal of a practical solution to make parking infrastructures not only smarter but also more sustainable, an essential goal for the cities of tomorrow.

II. Related Work

Our work is situated at the intersection of several research fields: smart parking technologies, IoT energy efficiency, and AI-based optimization.

A. Smart Parking Systems and Technologies

The evolution of smart parking systems has been primarily driven by advances in sensor technologies and IoT integration for real-time data collection. Comprehensive surveys by Rahman et al. [3] and Al-Turjman and Malekloo [2] provide a thorough overview of the architecture and challenges of these systems. Research has focused on various sensing modalities, such as ultrasonic or magnetic sensors embedded in wireless sensor networks [4], or more recent approaches using computer vision and deep learning to analyze overhead camera feeds [5]. While these systems provide valuable occupancy information, they primarily focus on the perception layer and often leave operational optimization aside.

B. Energy Efficiency in IoT and Smart City Applications

Energy optimization has become a critical research area in IoT systems, with particular emphasis on battery-powered devices and sensor networks. Anastasi et al. [6] provided a comprehensive survey of energy conservation techniques in wireless sensor networks, identifying adaptive duty cycling and intelligent sleep scheduling as key strategies for extending system lifetime. Similarly, Rault et al. [7] examined energy efficiency approaches, providing the theoretical foundation for adaptive power management that underpins our work on LED control.

Smart lighting systems represent the most relevant parallel to our problem. Work on smart street lighting, such as by Lau et al. [8] and Mahoor et al. [9], has proposed IoT-based frameworks that use sensors and adaptive control to achieve energy reductions of 30-60% while maintaining safety requirements. This research validates the potential for significant energy savings through intelligent illumination management in urban infrastructure, a principle we apply to the specific domain of parking facilities.

C. AI-based Optimization Approaches

To transition from simple information to optimization, research is turning to Artificial Intelligence.

Predictive Approaches (ML & DL): A significant body of literature focuses on occupancy prediction. Studies [10] have compared traditional statistical and Machine Learning models for availability forecasting. More recently, other works [11] demonstrated the effectiveness of Machine Learning-based approaches to predict parking space availability in IoT-enabled smart parking systems. However, these models remain fundamentally passive: they predict, but they do not actively control the infrastructure.

Decision-Making Approaches (RL): Reinforcement Learning is used for active decision-making across numerous smart city domains, such as intelligent traffic light control [12] and smart grid optimization [13]. Within the parking context, RL has been successfully applied to problems like dynamic pricing [14] or vehicle assignment to minimize user search time [15]. Nevertheless, our review reveals that the objective of these agents is rarely, if ever, the optimization of the infrastructure's own energy consumption. It is this research gap that our work aims to address.

III. System Modeling and Problem Formulation

A. System Architecture

We consider a smart parking facility with N parking spaces, where each space i is equipped with an occupancy sensor. The system operates continuously, with occupancy patterns varying according to time-dependent functions. Let $S_i(t)$ represent the occupancy state of space at time t:

$$S_i(t) = \begin{cases} 0 & \text{if space } i \text{ is occupied at time } t \\ 1 & \text{if space } i \text{ is available at time } t \end{cases} \tag{1}$$

The total occupancy at time t is defined as :

$$O(t) = \sum_{i=1}^{N} S_i(t) \tag{2}$$

The occupancy ratio is given by :

$$\rho(t) = \frac{O(t)}{N} \tag{3}$$

B. LED Management Strategies

In the traditional approach, each parking space maintains continuous LED illumination:

$$P_{LED}^{(1)}(i,t) = \begin{cases} P_{red} & \text{if } S_i(t) = 1 \\ P_{green} & \text{if } S_i(t) = 0 \end{cases} \tag{4}$$

where P_{red} and P_{green} represent the power consumption of red and green LEDs, respectively. For this analysis, we assume

$$P_{red} = P_{green} = P_{LED}$$

The total instantaneous power consumption is given for each strategy as follows:

Strategy 1 for the normal functioning:

$$P_{total}^{(1)}(t) = N \cdot P_{LED} \tag{5}$$

Strategy 2 (Adaptive Strategy), where we activate LEDs only for available spaces:

$$P_{LED}^{(2)}(i,t) = \begin{cases} 0 & \text{if } S_i(t) = 1 \\ P_{green} & \text{if } S_i(t) = 0 \end{cases} \tag{6}$$

The total instantaneous power consumption for Strategy 2 is:

$$P_{total}^{(2)}(t) = (N - O(t)) \cdot P_{LED} \tag{7}$$

Strategy 3 based on the intelligent RL: The guidance for a floor j is active $L_j(t) = 1$ if its occupancy $O_j(t)$ exceeds 50% of its capacity . The total power is the sum of the power of the active floors and is optimized by an RL-learned load-balancing policy:

$$P_{total}^{(3)}(t) = \sum_{j=1}^{3} (N_j - O_j(t)) \cdot P_{LED} \cdot L_j(t) \tag{8}$$

C. Energy Consumption Analysis

For Strategy 1 (Baseline):

$$E_1 = \int_0^T N \cdot P_{LED}\, dt = N \cdot P_{LED} \cdot T \tag{9}$$

For Strategy 2:

$$E_2 = \int_0^T N \cdot P_{LED} \cdot (1 - \rho(t))\, dt \tag{10}$$

For Strategy 3:

$$E_3 = \int_0^T \left(\sum_{j=1}^{3} (N_j - O_j(t)) \cdot P_{LED} \cdot L_j(t) \right) dt \tag{11}$$

The energy savings achieved by Strategy 3 compared to Strategy 1 is give as follows:

$$\Delta E = E_1 - E_3 = \int_0^T \left(N \cdot P_{LED} - P_3^{total}(t) \right) dt \tag{12}$$

The percentage energy savings is given by η as follows:

$$\eta = \frac{\Delta E}{E_1} \times 100\% = \left(1 - \frac{E_3}{N \cdot P_{LED} \cdot T} \right) \cdot 100\% \tag{13}$$

IV. Simulation framework and implementation

A. Simulation Parameters

The simulation was conducted using the following parameters:

TABLE I: Simulation Parameters

Parameter	Value
Number of parking spaces (N)	150
Number of Floors	3
Spaces per Floor (N_j)	50
LED power consumption (P_{LED})	5W
Simulation duration	*24 hours*
Time Resolution	*1 minute*
Occupancy Update Frequency	Real-time

B. Implementation Architecture

Our implementation is based on a distributed and modular architecture, decoupled by the MQTT protocol. It consists of three main software entities that interact to form our intelligent Digital Twin.

- The Intelligent Agent (RL-based): This is the brain of the system, developed in Python. It contains the learned decision-making policy and is responsible for observing the parking state and publishing optimal control commands.
- The Communication Layer (MQTT): The MQTT Broker acts as the nervous system, asynchronously managing all communication between components.
- The Digital Twin (Simulator and Dashboard): This entity serves a dual role. A Python script acts as the traffic simulator, generating arrival/departure events. In parallel, a web dashboard subscribes to the agent's commands and serves as the real-time visualization of the parking facility's state.

This architecture is not only flexible for simulation but is also directly transposable to a real physical system.

V. Result and Analysis

A. Occupancy Pattern Analysis

Figure 2 shows the occupancy pattern used in the simulation, representing typical weekday parking demand in an urban commercial area.

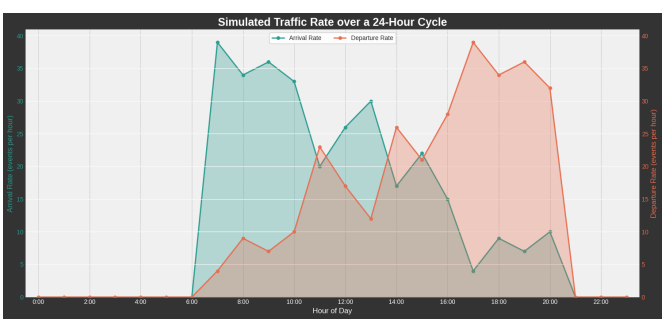


Fig. 2: Daily occupancy pattern for the parking facility

B. Validation of the RL Agent's Learning Process

The performance of Strategy 3 is entirely dependent on the agent's ability to learn an effective load-balancing policy. To validate this process, the agent was trained for over one hundred episodes. Figure 3 plots the evolution of the cumulative reward obtained by the agent at the end of each episode.

To better visualize the underlying trend, a 10-episode moving average is overlaid on the raw reward data.

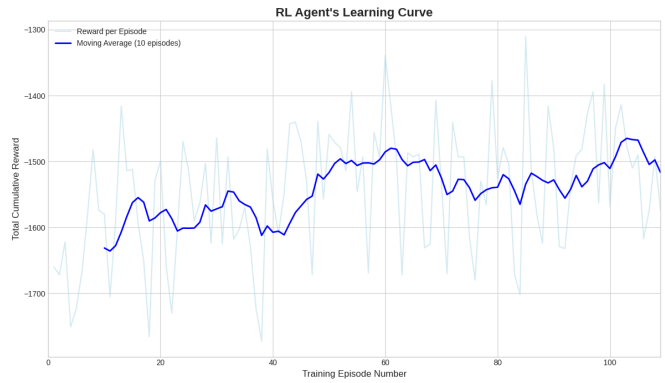


Fig. 3: RL Agent Training Convergence: Cumulative Reward per Episode.

The learning curve confirms successful policy acquisition. The moving average exhibits a distinct positive trend, progressing from an initial average reward of approximately -1650 to a stable plateau around -1450. This progression reflects the agent's transition from an initial, high-exploration phase (governed by a high ϵ value) to a subsequent exploitation phase where it consistently applies the learned optimal actions.

The convergence of the curve after approximately 80 episodes signifies that the agent has acquired a stable and effective load-balancing policy. The negative reward values are an expected outcome of our penalty-dominant reward function, which is designed to strongly discourage sub-optimal assignments. The successful convergence is a critical prerequisite, validating that the agent is capable of intelligent decision-making before we analyze its impact on energy consumption.

C. Instantaneous Power Consumption

Figure 4 illustrates the instantaneous power consumption for all three strategies throughout the 24-hour period.

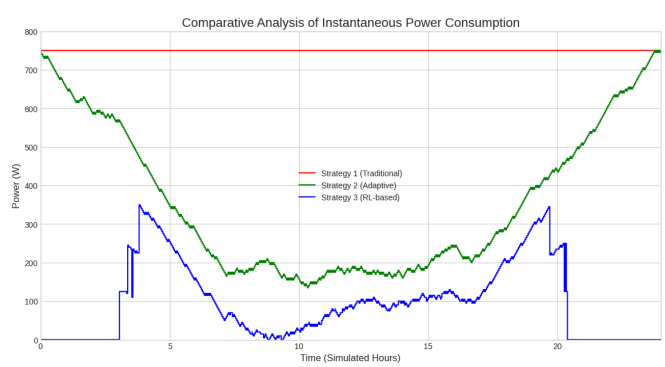


Fig. 4: Instantaneous power consumption comparison

D. Cumulative Energy Consumption

The cumulative energy consumption over 24 hours is shown in Figure 5.

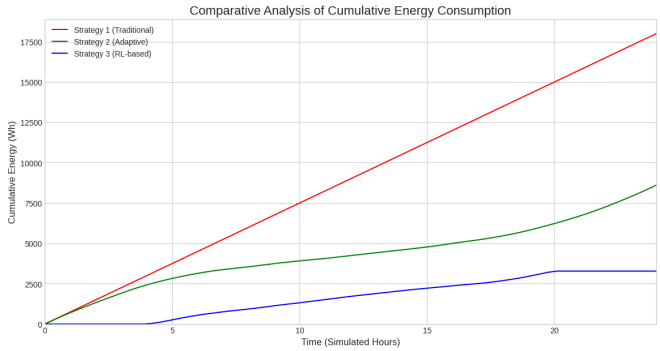


Fig. 5: Cumulative energy consumption over 24 hours

E. Energy Savings Analysis

Table II presents a detailed breakdown of energy consumption and savings for different time periods.

TABLE II: Energy Consumption and Savings Analysis

Time Period	Strategy 1 (kWh)	Strategy 2 (kWh)	Strategy 3 (kWh)	Savings S2 (%)	Savings S3 (%)
Night (00–06h)	4.477	3.133	0.537	**30%**	**88%**
Morning Rush (06–10h)	2.965	0.763	0.768	**74%**	**74%**
Peak Hours (10–16h)	4.477	1.075	1.047	**75%**	**76%**
Evening Peak (16–21h)	3.721	1.689	0.903	**54%**	**75%**
Late Evening (21–24h)	2.208	1.873	0.000	**15%**	**100%**
Total (24 hours)	**17.849**	**8.534**	**3.254**	**52%**	**81%**

The simulation was run for multiple iterations to ensure reproducibility. The quantitative analysis presented herein is based on a representative median run, with key performance metrics summarized below:

- Total Energy Savings: 81.8% compared to the traditional baseline.
- Peak Power Reduction: 750 W, corresponding to a 100% elimination of LED power consumption.
- Maximum Instantaneous Savings: 100%, achieved during low-activity periods.
- Minimum Instantaneous Savings: ∼53%, observed during peak hours.

VI. Discussion

A. Energy Efficiency Implications

The results demonstrate the exceptional energy savings potential of proactive, intelligent LED management. The proposed RL-based approach (Strategy 3) achieves:

- An 81.8% average daily energy reduction compared to the traditional baseline.
- 100% energy savings during extended low-activity periods (night, early morning, late evening).
- Maintained and optimized guidance functionality, activated only when and where it is needed.
- No compromise on the user experience, as spot assignment is managed seamlessly.

B. Economic Impact

Considering a typical commercial electricity rate of 0.12$/kWh, the annual cost savings for a single 150-space facility would be:

$$\text{Annual Savings} = 14.6\,\text{kWh/day} \times 365\,\text{days} \times 0.12\,\$/\text{kWh}$$

$$= 639.48\,\$$$

For large-scale deployments across urban areas (e.g., university campuses, airports, shopping centers), these savings become highly substantial.

C. Environmental Benefits

The CO_2 emission reduction, assuming an average grid emission factor of 0.5 kg CO_2/kWh, is given by the $ACO2R$ (Annual CO_2 Reduction) as follows:

$$ACO2R = 14.6\,\text{kWh/day} \times 365\,\text{days} \times 0.5\,\frac{\text{kg CO}_2}{\text{kWh}}$$

$$= 2664.5\,\text{kg CO}_2 \approx 2.7\,\text{tonnes of CO}_2 \text{ per facility, per year.}$$

VII. Future Work

Several research directions emerge from this study:

A. Multi-Objective Reinforcement Learning

Evolving the agent's reward function into a multi-objective framework could enable policies that concurrently optimize energy consumption, minimize user walking distance, and reduce vehicle wait times.

B. Dynamic Control Policies

A more advanced agent could learn not only the assignment strategy but also dynamic, context-aware operational rules, such as adjusting the 50% activation threshold based on real-time traffic intensity or time of day.

C. Transfer Learning for Scalability

Investigating transfer learning could allow a pre-trained agent to be rapidly adapted to new parking facilities with minimal data, significantly reducing the training cost and complexity of large-scale, multi-facility deployments.

D. City-Scale Multi-Agent Systems (MAS)

Expanding the single-agent concept to a multi-agent system where agents from different facilities communicate and coordinate would allow for proactive, city-level traffic management, diverting vehicles from congested areas to optimize urban mobility on a broader scale.

VIII. Conclusion

This research has demonstrated the substantial potential of reinforcement learning-based energy management for smart parking systems, achieving an remarkable 81.8% reduction in average daily energy consumption compared to traditional approaches. The strategy optimizes energy usage through temporal optimization—strategically delaying system activation and accelerating deactivation during low-activity periods—while maintaining operational efficiency. Economic analysis reveals significant annual cost savings of approximately $640 per facility alongside environmental benefits equivalent to reducing 2.7 tonnes of CO_2 emissions per year per facility.

The study establishes a novel hybrid modeling framework that effectively couples stochastic decision policies with deterministic operational rules, validated through a comprehensive Digital Twin-based simulation environment. This approach quantitatively demonstrates the superiority of proactive control strategies that shape system states over reactive approaches that merely adapt to existing conditions. Importantly, the solution is primarily software-based, enabling cost-effective integration into existing smart parking infrastructure without requiring extensive hardware modifications.

These findings have broader implications for smart city development, supporting a paradigm shift toward proactive, intelligent IoT systems. The quantified economic and environmental benefits provide strong justification for integrating such intelligent energy management approaches into future urban planning policies and technology standards, contributing to the development of more sustainable smart cities.

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